

Rohit U Bogulla

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Summary

Software Engineer with 5+ years of production experience building agentic AI systems, real-time data pipelines, and cloud-native backend services across enterprise and research environments. Strong programming skills in Python, Java, C/C++, and Go with hands-on expertise in LangChain, LangGraph, Apache Kafka, Spark Structured Streaming, Kubernetes, and AWS, shipping production systems spanning LLM-powered research platforms, NLP pipelines, enterprise billing infrastructure, and microservice deployment frameworks.

Experience

AI Engineer – *University of Florida*

May 2025 – Present

- Designed production agentic AI platform using LangChain, LangGraph, and GPT-4 with RAG-based retrieval and NLP pipelines querying across 10+ biodiversity databases to resolve natural language queries at 200ms latency serving 1000+ daily users via <https://ichatbio.org>
- Built cloud-native infrastructure on Docker and Kubernetes with automated DeepEval test coverage, Pydantic validation, and production monitoring for latency, error rates, and resource utilization

Software Engineer – *Oracle*

Aug 2022 – Jul 2024

- Developed and maintained features across Oracle BRM in C and C++ spanning Billing, Subscription Manager, Balances, Rating, and Pipeline Manager components processing 10M+ events daily serving 50M+ subscriber accounts
- Delivered 20% performance improvement in C/C++ transaction-processing pipelines; architected cloud-native data partitioning for 50M+ subscriber accounts; built dynamic pricing engine adopted by 3+ tier-1 telecom operators via Java Spring Boot REST APIs
- Established CI/CD pipelines reducing deployment time by 50%; drove memory management improvements and SQL optimization; mentored junior engineers on Oracle BRM internals and SDLC best practices

Associate Software Engineer – *Oracle*

Sep 2020 – Aug 2022

- Built and deployed production REST APIs for Oracle DX4C (Digital Experience for Communications), integrating Pricing Design Center with downstream billing systems via Java/Helidon services
- Reduced API latency by 30% through query optimization and strategic caching; authored technical specifications guiding cross-team API development and cost-optimization strategies

Software Engineer Intern – *GoTo Technologies (LogMeIn)*

Jan 2020 – Sep 2020

- Built Nucleus, a rapid deployment framework automating microservice and Lambda deployment to AWS (ECS, EKS, Route53, S3) with Jenkins CI/CD, Docker, and Kubernetes (Helm); adopted by 2 major LogMeIn products reducing deployment time from hours to minutes
- Integrated Istio service mesh for traffic management and observability; configured Wavefront dashboards alongside Prometheus tracking deployment health, latency, and failure signals across the Nucleus cluster

Projects

Comic/Manga Page Interpretation via Vision-Language Models – *arxiv*

Jan 2025 – May 2025

- Engineered end-to-end pipeline for making comics and manga accessible to visually impaired users via screen reader integration; developed prompt engineering strategies for VLMs improving page-level comprehension accuracy by 40%
- Benchmarked 6 Vision-Language Models (MiniCPM, Qwen 2.5, LLaVA 1.6, InstructBLIP, CogVLM, Idefics3) on hallucination metrics; built evaluation framework analyzing VLM response quality across diverse comic genres using Python, PyTorch, and HuggingFace Transformers

CrimeLens – End-to-End NLP Data Pipeline

Sep 2025 – Dec 2025

- Built an end-to-end NLP-to-GNN pipeline (spaCy + PyTorch Geometric) that converts free-text crime scene descriptions into structured entity-event graphs and classifies crime type via a 3-layer Residual GCN, achieving 92% test accuracy across 8 categories on a 1,500-scene synthetic dataset; added GNNExplainer-style interpretability and a Streamlit dashboard for interactive prediction and graph visualization.

JobSignal – MLOps with Kafka and Spark

May 2026

- Built a real-time email intelligence pipeline using Kafka, Spark Structured Streaming, and spaCy/DistilBERT NLP; integrated MLflow for experiment tracking and Prometheus + Grafana for live observability; containerized with Docker and deployed CI via GitHub Actions

Skills

Languages: Python, Java, C, C++, Go, TypeScript, React, JavaScript, SQL

Agentic AI & ML: LangChain, LangGraph, RAG, GPT-4, HuggingFace Transformers, PyTorch, PyTorch Geometric, spaCy, DistilBERT, DeepEval, prompt engineering, NLP, scikit-learn

Data & Streaming: Apache Kafka, Spark, MLflow, PostgreSQL, MySQL, Redis (NoSQL), ChromaDB

Cloud & DevOps: AWS (ECS, EKS, S3, Route53), Docker, Kubernetes (Helm), Jenkins, GitHub Actions, CI/CD, Linux
Observability & QA: DeepEval, Prometheus, Grafana, Wavefront, Istio service mesh, CI/CD quality gates

Education

University of Florida	Gainesville, FL
<i>M.S. in Computer Science GPA: 4.00 / 4.00</i>	<i>May 2026</i>